R7.1: smooth GHS descended curve for m >= 2, and R6.9′s relation rate on it

2026-10-25. Stacked on PR #2502 (R6.9). Source: `RESULTS.md` (same directory), raw data `relrate_descended.json`.

Result. The library builds the smooth model of the GHS descended curve C/k and $\Phi : E(K) \to \mathrm{Jac}(C)(k)$ for $m \geq 2$. On all 13 measured curves: genus confirmed by N_{g+1} predicted from $N_1..N_g$; Φ a homomorphism (260 of 260 checks); $\#E \cdot \Phi = 0$ and $\#J \cdot \Phi = 0$ (156 checks, 0 failures); the degree-$2n$ Frobenius relation holds with a trace-perturbed negative control at 0. At N = 12, $(l, n) = (2, 6)$, $g = 8$, $q = 4$ (6 curves) the measured fully-split rate is 0.00064 to 0.00261 per step, 0.87 to 1.23 times the enumerated split fraction, and 26 to 105 times Gaudry's $1/g!$ (recalled). R6.9′s random stand-ins gave 585 and 858 times; the real descended Jacobians have only 5 F_4-points, hence 60 split divisors. The $(3, 4)$, $g = 8$ run did not finish (stopped, cause not isolated). No claim about any deployed curve or any cost against rho.

Relation rate

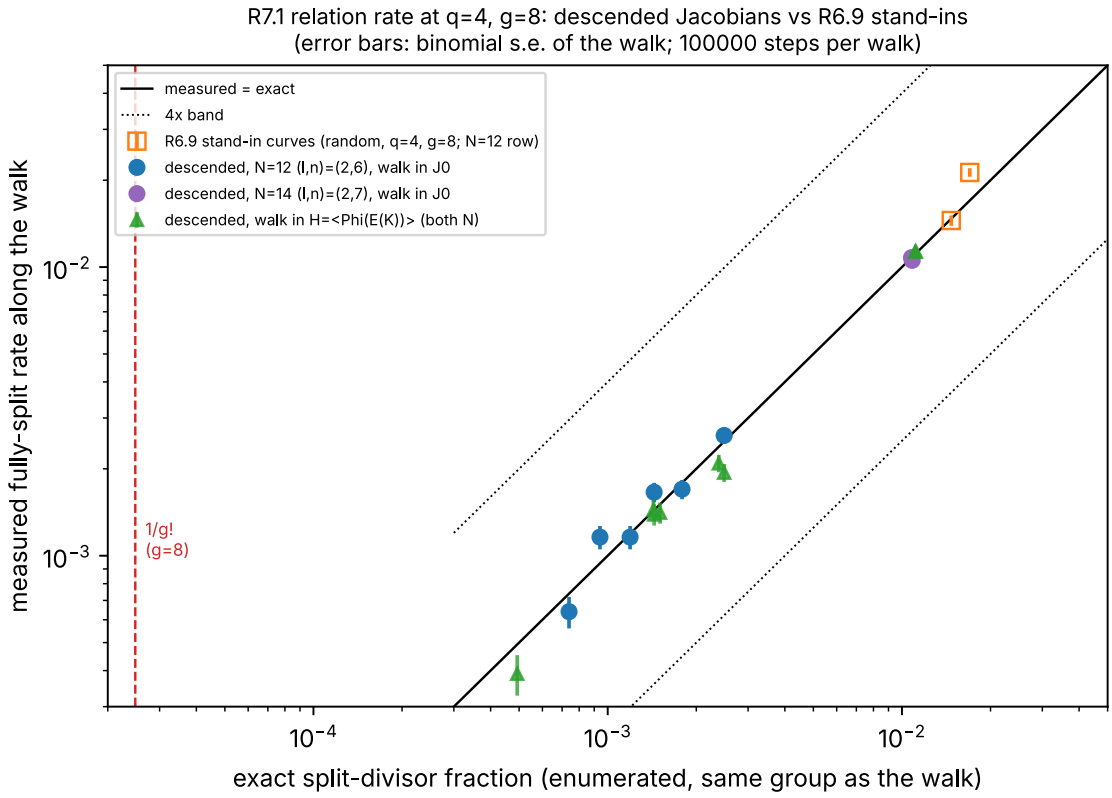


Figure 1: Measured fully-split rate against the exactly enumerated split fraction of the walked group, $q = 4$, $g = 8$. 100000 steps per walk; error bars are binomial standard errors of an assumed-independent walk. Orange: R6.9′s stand-in curves (recorded values). Source: `relrate_descended.json`, `figure_data.tsv`.

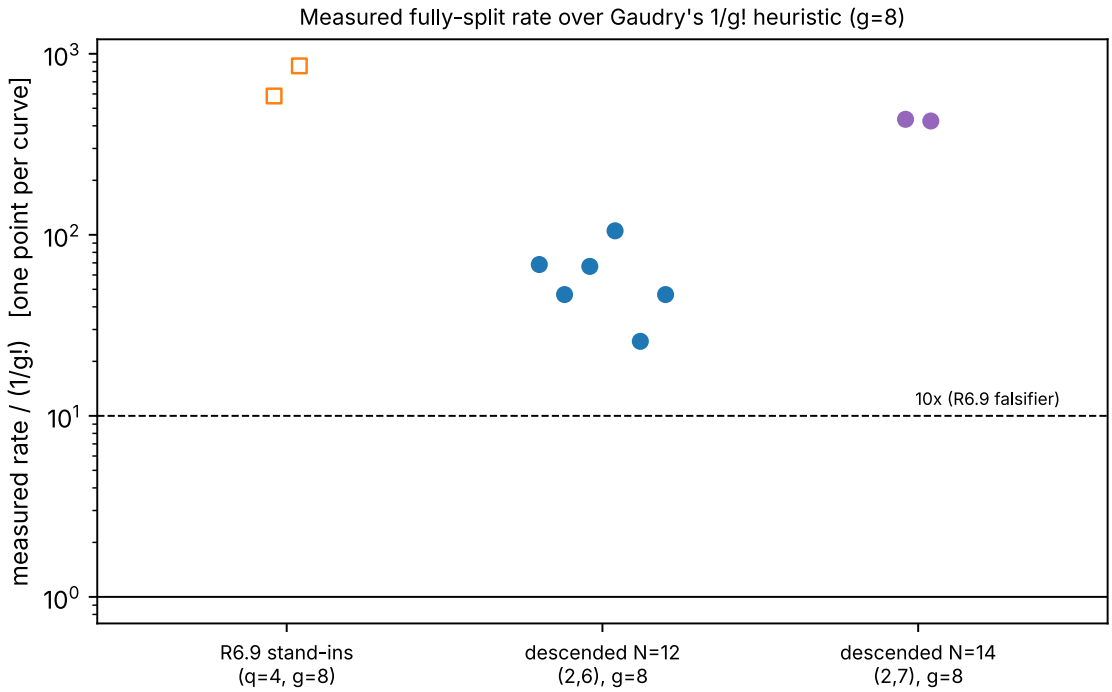


Figure 2: Measured rate over $1/g!$ per curve, $g = 8$, walk in J_0 (derived from the recorded rates). Dashed: R6.9′s 10x falsifier.

Verification summary

Check	Result
`ghs_smooth` library tests	5 passed, 0 failed
Genus vs point counts (N_{g+1} predicted)	13 of 13
Φ homomorphism	260 ok, 0 failed
Order checks	156, 0 failures
Frobenius relation gcd degree	$2n$ (or $2g$); control 0
Closed-form split count = enumeration in J_0	13 of 13

Not settled

Larger N or q; the $q = 8$, $g = 8$ instance; odd-genus t-model beyond one curve; $a \notin k$; the factor-base decomposition step; any cost comparison with rho.